A Shared LSTM Forecast for 3-D UAV Planning with Coupled MPC–DWA in Dynamic Obstacle Fields

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***Abstract*—First, a UAV must avoid moving obstacles in three dimensions even when turns, acceleration and pauses make their motion difficult to predict. Second, the method couples model predictive control (MPC) and the dynamic window approach (DWA) through one causal long short-term memory (LSTM) forecast. MPC uses it to optimize a route; DWA screens reachable actions and bounds MPC's first command. Then, in a frozen 500-scene comparison with a 1-s DWA screen, shared LSTM achieved collision-free goal arrival in 936/1,500 runs (62.4%), versus 836/1,500 (55.7%) when only DWA received LSTM. The paired difference was 6.7 percentage points (95% scene-cluster confidence interval, 4.2–9.2). In a separate 100-scene exploratory evaluation with a 3-s DWA screen, sharing reached 73.7%; its contrast with DWA-only LSTM was inconclusive. Standalone MPC-CV reached 92.0%, the highest rate in that evaluation. Finally, the simulations show a gain from forecast sharing under the frozen 1-s DWA protocol, while standalone MPC-CV led the separate 3-s evaluation.**

***Index Terms*—UAV, 3D path planning, dynamic obstacles, LSTM, model predictive control, dynamic window approach**

I. Introduction

Unmanned aerial vehicles (UAVs) must avoid obstacles that move during flight. Turns, acceleration and pauses make their recent positions unreliable guides to what happens next. Model predictive control (MPC) optimizes a short future route; the dynamic window approach (DWA) chooses a feasible action for the next step [1], [2]. Recent UAV planners also use predicted obstacle motion in MPC [3], [4]. We test whether one causal long short-term memory (LSTM) forecast, shared by MPC and DWA, improves the coupled planner relative to one-layer forecast input, constant-velocity (CV) prediction and standalone controllers. We also compare with a project-local latest-position-hold reconstruction of the MPC–DWA baseline, because the source paper does not specify an executable dynamic-obstacle predictor [5]. The shared forecast connects route optimization to DWA's reachable-action screening and local safety gates.

II. Problem Formulation

We model the UAV as a collision sphere in a bounded three-dimensional workspace with a start, goal and moving spherical obstacles. Each obstacle has an observed center, radius and time-varying path; future ground-truth positions are hidden from the planner. At each update, it chooses a dynamically feasible action to reach the goal without intersecting an obstacle along the simulated motion. Figure 1 shows the scene geometry; recorded planner paths are compared separately.

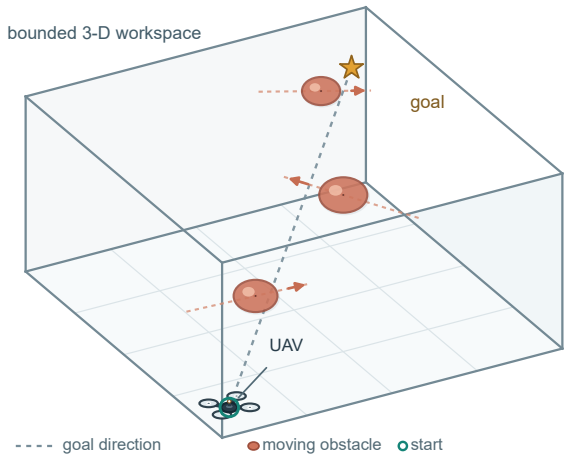


Figure 1. Problem geometry: the quadrotor is centered at the start and moves toward the goal among spherical obstacles following prescribed paths (arrows). The dashed line marks the direct start-to-goal direction, not a simulated route; planner trajectories appear separately in Figure 2.

III. Planning Method

With the task geometry defined, we use one causal obstacle forecast to connect route optimization with locally reachable action selection.

A. Core equations

The vehicle model, causal obstacle forecasts, MPC objective and DWA coupling are collected here. The first four relations update position, speed and attitude; the final relation defines signed surface clearance:

$$\begin{aligned}
\mathbf{p}_{t+1} &= \mathbf{p}_t + \Delta t\, v_t[\cos\theta_t\cos\psi_t, \cos\theta_t\sin\psi_t, \sin\theta_t]^{\mathsf{T}},\\
v_{t+1} &= v_t + \frac{\Delta t}{\tau_v}(v_{c,t} - v_t),\\
\theta_{t+1} &= \theta_t + \Delta t\,\omega_{\theta,t}, \quad \psi_{t+1} = \psi_t + \Delta t\,\omega_{\psi,t},\\
\hat{d}_{j,k} &= \|\mathbf{p}_{t+k|t} - \hat{\mathbf{o}}_{j,t+k|t}\|_2 - r_u - r_j.
\end{aligned} \tag{1}$$

In (1), t and k denote current and future steps; $j = 1, \ldots, M$ indexes the M obstacles. The UAV position, speed, pitch and yaw are $\mathbf{p}$, v, θ and ψ; commands v_c, ω_θ and ω_ψ set speed and angular rates. The update interval is Δt, τ_v is the speed-response time, r_u is the UAV radius and r_j is obstacle j's radius. Thus $\hat{d}_{j,k}$ is signed surface clearance: positive means separated, zero contact and negative overlap.

The constant-velocity (CV) baselines extrapolate from the two latest observed positions; at startup, when only one position is available, the initial velocity is set to zero. A separate latest-position-hold baseline keeps the latest observed center fixed over the forecast horizon. The causal LSTM [6] maps ten consecutive observations to thirty three-dimensional offsets from the latest position:

$$
\begin{aligned}
\hat{\mathbf{v}}_{j,t} &= (\mathbf{o}^{\mathrm{obs}}_{j,t} - \mathbf{o}^{\mathrm{obs}}_{j,t-1})/\Delta t, \\
\hat{\mathbf{o}}^{\mathrm{CV}}_{j,t+k|t} &= \mathbf{o}^{\mathrm{obs}}_{j,t} + k\Delta t\, \hat{\mathbf{v}}_{j,t}, \\
\mathbf{q}_{j,s} &= [(\mathbf{o}^{\mathrm{obs}}_{j,s})^{\mathsf{T}}, (\mathbf{v}^{\mathrm{obs}}_{j,s})^{\mathsf{T}}, \cos\chi_{j,s}, \sin\chi_{j,s}]^{\mathsf{T}}, \\
\hat{\boldsymbol{\delta}}_{j,t+1:t+30|t} &= f_\Theta(\mathbf{q}_{j,t-9:t}), \\
\hat{\mathbf{o}}^{\mathrm{LSTM}}_{j,t+k|t} &= \mathbf{o}^{\mathrm{obs}}_{j,t} + \hat{\boldsymbol{\delta}}_{j,t+k|t}.
\end{aligned}
\tag{2}
$$

In (2), s indexes history samples; $\mathbf{q}$ contains eight features (3-D position, 3-D velocity, and sine/cosine of heading χ), and f_Θ is the trained LSTM with learned parameters Θ. Here $\mathbf{v}^{\mathrm{obs}}_{j,s}$ is the simulator-provided obstacle velocity, distinct from the finite-difference CV estimate $\hat{\mathbf{v}}_{j,t}$; $\chi_{j,s}$ is obstacle heading (obstacle yaw), not UAV yaw ψ_t. Ten observations span 0.9 s; the LSTM predicts 30 offsets, each anchored to the latest observed center rather than cumulatively summed. At startup, the earliest available state fills missing history. The forecast is causal: at planning time t, the network receives only observations through t. The benchmark compares configured controllers: LSTM receives simulated position, velocity and heading, whereas CV extrapolates the latest two observed positions. Training details are in Supplementary Methods.

In the shared-LSTM arm, one time-aligned forecast is supplied to both planners so MPC's route and DWA's screened actions anticipate the same obstacle motion. This is designed to align long-horizon route optimization with the near-term action check, rather than plan against two different obstacle futures. MPC uses the forecast to optimize its route, while DWA uses corresponding near-term positions to screen reachable commands. Because MPC represents obstacle safety with soft penalties, DWA also enforces explicit clearance, corridor and braking gates. The accepted DWA actions set per-coordinate bounds on MPC's first command in (3). In the historical DWA-only ablation, LSTM informs DWA while MPC holds the latest observed obstacle positions. The MPC objective and coupling constraints are:

$$
\begin{aligned}
\mathbf{a} &= \mathbf{g} - \mathbf{s}, \\
\eta_{\mathbf{a}} &= \mathbf{a}^{\mathsf{T}}\mathbf{a} + 10^{-12}\,\mathrm{m}^2, \\
P_{\mathbf{a}} &= I_3 - \frac{\mathbf{a}\mathbf{a}^{\mathsf{T}}}{\eta_{\mathbf{a}}}, \\
\mathbf{e}^{\perp}_{t+k|t} &= P_{\mathbf{a}}(\mathbf{p}_{t+k|t} - \mathbf{s}), \\
J_r &= \sum_{k=1}^{30} \|\mathbf{e}^{\perp}_{t+k|t}\|_2^2, \\
\min_{\mathbf{u}_1,\ldots,\mathbf{u}_{30}} \quad J_{\mathrm{MPC}} &= \sum_{k=0}^{29} w_p \|\mathbf{p}_{t+k|t} - \mathbf{g}\|_2^2 \\
&\quad + \sum_{k=1}^{30} w_u \|\mathbf{u}_k\|_2^2 \\
&\quad + w_f \|\mathbf{p}_{t+30|t} - \mathbf{g}\|_2^2 + w_r J_r \\
&\quad + w_{\Delta u} \sum_{k=2}^{30} \|\mathbf{u}_k - \mathbf{u}_{k-1}\|_2^2 \\
&\quad + \sum_{k=1}^{30}\sum_{j=1}^{M} w_o [-\hat{d}_{j,k}]_+^2 \\
&\quad + \sum_{k=1}^{30}\sum_{j=1}^{M} w_s [d_{\mathrm{soft}} - \hat{d}_{j,k}]_+^2, \\
&\text{subject to } \mathbf{x}_{t|t} = \mathbf{x}_t, \\
&\mathbf{x}_{t+k|t} = f(\mathbf{x}_{t+k-1|t}, \mathbf{u}_k), \\
&\qquad k = 1, \ldots, 30, \\
&\mathbf{u}_k \in [0, 2.0] \times [-0.8, 0.8]^2, \\
&\mathbf{u}_1 \in \mathcal{U}_t \quad \text{when } \mathcal{A}_t \neq \varnothing, \\
&|\mathbf{u}_k - \mathbf{u}_{k-1}| \preceq \mathbf{a}_u \Delta t, \\
&\qquad k = 1, \ldots, 30.
\end{aligned}
\tag{3}
$$

Here $\mathbf{x}_t = [\mathbf{p}_t^{\mathsf{T}}, \theta_t, \psi_t, v_t]^{\mathsf{T}}$ is the position–pitch–yaw–speed state; f is defined by the first four state-update relations in (1). The MPC command is $\mathbf{u}_k = [v_{c,k}, \omega_{\theta,k}, \omega_{\psi,k}]^{\mathsf{T}}$; $\mathbf{u}_0$ is the last applied DWA command and $\mathbf{a}_u$ gives componentwise rate limits. The physical input box means commanded speed is 0–2.0 m s^{-1} and each angular rate is limited to ± 0.8 rad s^{-1}. In the rate constraint, absolute value and $\preceq$ act componentwise, so each command component may change by at most its corresponding limit in $\mathbf{a}_u$ over Δt. The start and goal are $\mathbf{s}, \mathbf{g}$, and $\mathbf{a} = \mathbf{g} - \mathbf{s}$ is the route vector. The scalar $\eta_{\mathbf{a}}$ is its squared length plus the implementation's $10^{-12}\,\mathrm{m}^2$ numerical safeguard; $P_{\mathbf{a}}$ is the resulting regularized cross-track operator, and J_r penalizes the residual distance from the direct line. I_3 is the three-dimensional identity. The index k runs over the prediction horizon and $\hat{d}_{j,k}$ is signed clearance for obstacle j; MPC adds a $10^{-12}\,\mathrm{m}^2$ numerical smoothing term when evaluating distance, whereas DWA and swept-collision checks use the exact norm. Weights $w_p, w_u, w_f, w_r, w_{\Delta u}, w_o, w_s$ penalize, in order, stage goal error, control effort, terminal goal error, route deviation, command changes, overlap and clearance below $d_{\mathrm{soft}} = 0.35$ m. The positive-part operator

is $[x]_+ = \max(0, x)$, so a penalty activates only when its clearance threshold is violated. The implementation squares native command values without cross-channel normalization, making the effort term a heuristic scalarization rather than a dimensionally homogeneous physical cost; numerical weights and command limits are given in Supplementary Methods. The MPC solver imposes no separate state-box bounds. When the normal safe set $\mathcal{A}_t$ is nonempty, accepted DWA actions define a coordinate-wise box $\mathcal{U}_t$ for MPC's first input; the box may contain actions DWA did not screen and is not itself a safety certificate for every enclosed action.

Trust score τ combines predicted safety, goal progress, agreement with the previous MPC guide, and control smoothness. It sets the sample budget and guided/exploratory split. Components and weights lie in $[0, 1]$; weights sum to one.

$$\begin{aligned} \tau &= \sigma_s \tau_{\mathrm{sa}} + \sigma_g \tau_g \\ &\quad + \sigma_t \tau_{\mathrm{ref}} + \sigma_m \tau_{\mathrm{sm}}, \\ \Delta_p &= \|\mathbf{p}_t - \mathbf{p}_{\mathrm{ref},1|t-1}\|_2, \\ \tau_{\mathrm{ref}} &= \exp(-\alpha_T \Delta_p), \\ B &= \mathrm{round}[300(1 + \alpha e^{-\beta\tau})], \\ B_g &= \mathrm{round}(B\tau), \quad B_e = B - B_g. \end{aligned} \tag{4}$$

In (4), $\sigma_s, \sigma_g, \sigma_t, \sigma_m$ weight safety, goal progress, reference tracking and smoothness; they sum to one, and σ_t denotes tracking rather than time. The reference position $\mathbf{p}_{\mathrm{ref},1|t-1}$ is the first predicted point on the previous MPC route, so Δ_p measures the UAV's distance from that guide. By definition, τ_{ref} decreases with this distance; the implementation names this tracking score τ_t. At startup, scores use the initial MPC solution; later they use the shifted route and controls. The four scores measure clearance-based safety, route progress, agreement with the previous route's first point, and control smoothness; their full definitions are in Supplementary Methods S1.2. The historical benchmark retained inherited weights $[0.35, 0.30, 0.15, 0.20]$, not tuned on its 500 scenes. A separate 4,320-episode development sweep (six vectors, five integrated branches, 48 scenes and three matched seeds) selected $[0.15, 0.15, 0.15, 0.55]$, with equal-arm macro-average collision-free arrival of 0.8042 (branch rates in Supplementary Methods S1.2). Because the same runs selected the vector, this score is descriptive, not held-out evidence: the historical benchmark retained its inherited vector, while the current 100-scene evaluation uses the selected vector. Its $\alpha_T = 3\ \mathrm{m}^{-1}$ and dimensionless $\beta = 3$ settings remained fixed and were not selected by the incomplete Stage-2 grid. B is split into MPC-guided (B_g) and exploratory (B_e) samples, and the clipped MPC command is also tested. The dimensionless budget gain $\alpha = 1$ is distinct from α_T; higher trust reduces exploration and sample count, 300 is the base budget, and β sets sensitivity.

DWA rejects candidate b unless its rollout meets the clearance, flight-corridor and braking checks. It ranks the remaining actions and uses their per-coordinate ranges to bound MPC's first control:

$$\begin{aligned} \mathcal{A}_t &= \{b : d_{\min}^{(b)} > 0.25\ \mathrm{m},\ \mathcal{C}_b = 1, \\ &\qquad (v_{\mathrm{end}}^{(b)})^2 \le 2 a_{\max} d_{\min}^{(b)}\}, \\ S_b &= w_p^{\mathrm{DWA}} S_{p,b} + w_h^{\mathrm{DWA}} S_{h,b} \\ &\quad + w_v^{\mathrm{DWA}} S_{v,b} + w_o^{\mathrm{DWA}} S_{o,b} + w_z S_{z,b}, \\ b^* &= \arg\max_{b \in \mathcal{A}_t} S_b, \\ u_{i,t}^- &= \max(u_i^{\min}, \min_{b \in \mathcal{A}_t} u_i^{(b)} - \delta), \\ u_{i,t}^+ &= \min(u_i^{\max}, \max_{b \in \mathcal{A}_t} u_i^{(b)} + \delta), \\ \mathcal{U}_t &= \prod_{i=1}^{3} [u_{i,t}^-, u_{i,t}^+]. \end{aligned} \tag{5}$$

In (5), $d_{\min}^{(b)}$ is candidate b's smallest discrete surface clearance over all obstacles and sampled rollout states; $\mathcal{C}_b = 1$ means its rollout stays inside the flight corridor. A candidate is retained only when that minimum exceeds 0.25 m and the stopping-distance test passes; $v_{\mathrm{end}}^{(b)}$ and $a_{\max}$ are terminal speed and available deceleration. Scores S_p, S_h, S_v, S_o measure MPC-endpoint agreement, heading agreement, speed tracking with a forward-speed reward, and normalized obstacle clearance; S_z penalizes vertical-profile deviation. Trust-dependent weights on the first four scores sum to one; $w_z = 0.25$ is fixed and not renormalized. The margin $\delta = 0.1$ is applied separately to each command coordinate (not as one shared physical distance) to expand the screened candidates' range; i indexes speed, pitch rate and yaw rate, and $u_i^{\min}, u_i^{\max}$ are physical limits. The top-scoring safe candidate is normally applied. A yaw-rate correction is triggered when the selected rollout's minimum predicted clearance exceeds 1.0 m while the UAV's current horizontal cross-track offset from the start–goal line exceeds 0.5 m; the adjusted command is re-screened by the same clearance, corridor and braking tests, and the original safe command is restored if it fails. Collision outcomes are independently checked on continuous swept paths. If the normal set is empty, the emergency branch is screened and the run stops if none passes. Full score formulas and settings are in Supplementary Methods.

B. How the shared forecast is used

At each 0.1-s update, the shared arm sends the same causal 3-s LSTM forecast to MPC and DWA. MPC optimizes a 30-step route; DWA screens reachable 30-step actions against that forecast, and its passing actions bound MPC's first command on re-solve. The updated MPC route guides the next DWA ranking, while the screened DWA action is executed. In the one-layer ablations, only the named layer receives LSTM predictions and the other uses CV; independent DWA and MPC use CV, and the no-forecast comparator holds the latest obstacle positions. The executed cycle and fallback are detailed in Supplementary Methods S1.4 and summarized in Algorithm S1.

IV. Simulation and Results

Under the frozen 1-s DWA/3-s MPC protocol and inherited weights, 500 scenes (three matched seeds each) compared shared LSTM with DWA-only LSTM. DWA received LSTM forecasts in both arms; only shared LSTM also supplied them to MPC. Safe arrival was 936/1,500 (62.4%) versus 836/1,500 (55.7%; +6.7 percentage points, 95% scene-cluster CI, 4.2–9.2). These 3,000 records belong to the 7,500-run five-controller benchmark, not an additional sample; full outcomes and protocol are in Supplementary Methods S4.

Separately, an exploratory seven-arm comparison tested 100 scenes (three matched seeds; 300 runs per arm) using 0.1-s updates, 30-step (3-s) MPC/DWA horizons and a 650-step limit. Arms were standalone DWA-CV and MPC-CV, coupled MPC–DWA-CV, LSTM-to-MPC-only, LSTM-to-DWA-only, shared LSTM and latest-position-hold MPC–DWA. Safe arrival required goal tolerance without swept-path contact. Seeds were averaged within scene; 20,000 scene-cluster bootstrap resamples produced simultaneous max-—t— intervals for six shared-versus-comparator contrasts. The scene bank was distinct from the 500-scene benchmark.

A matched route from the five-controller benchmark (scene 404, seed 2) is shown in Fig. 2. Shared LSTM arrived safely, whereas the other four arms did not; aggregate rates use all matched runs.

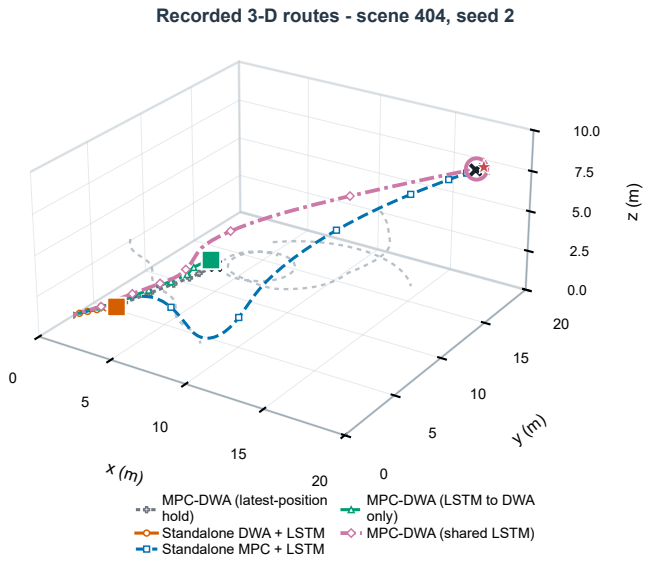


Figure 2. Recorded simulated 3D paths from the five-controller, 500-scene benchmark (scene 404, sampling seed 2). Shared LSTM achieved collision-free goal arrival. The reconstructed MPC–DWA baseline and standalone MPC + LSTM had swept contact; standalone DWA + LSTM and MPC–DWA with LSTM supplied only to DWA stopped before arrival. Each curve uses saved positions through its logged terminal state, with no smoothing, interpolation or extension. Triangle/star: start/goal; circle: safe arrival; square: no-contact stop; X: a run with swept contact, not its contact location; dashed lines: obstacle tracks. The outcome-selected matched case illustrates route geometry; aggregate results use all runs.

In the 3-s evaluation, shared LSTM reached safely in 221/300 runs (73.7%), 16.3 points above position-hold (simultaneous 95% CI, 5.1–27.6) but 18.3 below MPC-CV (−29.6 to −7.0). Intervals versus DWA-CV, coupled CV and both one-layer LSTM arms, including DWA-only, included zero. Standalone MPC-CV ranked highest; shared LSTM exceeded position-hold, while its DWA-only contrast was inconclusive.

Table 1
Outcome summary for the exploratory simulation ($n = 300$ episodes per arm). Safe arrival requires reaching the goal without swept-path contact. Full paired intervals are in Supplementary Table S12.

Controller	Safe-goal rate (%)	Swept contacts
Standalone DWA-CV	78.0	0
Standalone MPC-CV	92.0	15
Coupled MPC–DWA-CV	68.0	0
MPC–DWA (LSTM to MPC only)	73.0	0
MPC–DWA (LSTM to DWA only)	74.7	0
MPC–DWA (shared LSTM)	**73.7**	**0**
MPC–DWA (latest-position hold)	57.3	50

V. Discussion

The two protocols indicate a protocol-specific benefit: forecast sharing raised safe arrival in the frozen 1-s screen comparison, but not in the 3-s exploratory comparison; standalone MPC-CV had the highest rate there. Because scene banks, trust weights and horizons also differed, these studies do not isolate the source of that difference. LSTM uses simulator-recorded obstacle velocity and heading, whereas CV estimates velocity from positions, so the contrast concerns configured input pipelines as well as predictors. Prescribed tracks, point forecasts and no flight tests limit generalization.

VI. Conclusion

In simulation, shared LSTM increased collision-free arrival by 6.7 percentage points over DWA-only input in the frozen 500-scene, 1-s DWA comparison. In the separate 3-s evaluation, standalone MPC-CV had the highest rate, and the shared-versus-DWA-only contrast was inconclusive.

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